

Model selection and goodness-of-fit testing: the basics

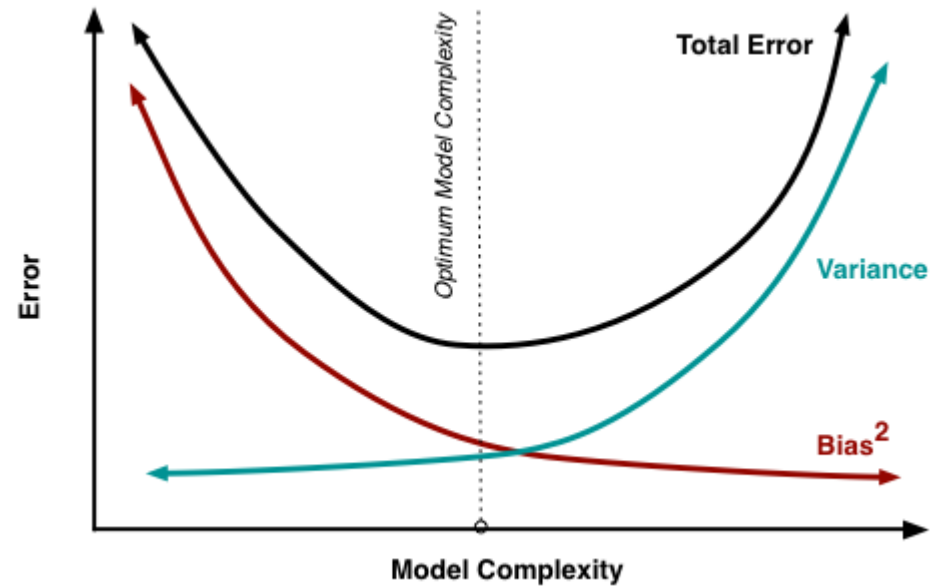
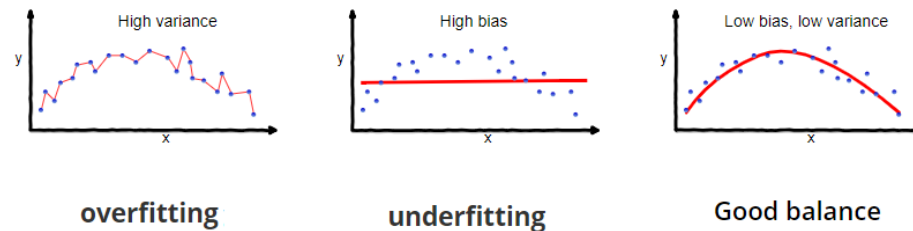


Table 1. Models examining effects of various covariates on detection probabilities of indicated breeding pairs of waterfowl in North Dakota, USA (from Pagano and Arnold 2009). I added single parameters assuming an additive effect to the base model, which included $K = 16$ parameters (8 species $\times$ 2 observers). Three of these covariates were considered biologically feasible (total ducks, vegetative cover, and cover type), 6 were not (random 5, 4, 8, and 1; not Sunday, Monday, or Wednesday; and last duck seen a mallard), and I excluded 6 additional nonsense or random variables ($\Delta AIC = 0.64$ –2.00) from presentation. I evaluated all models compared to the base model using Akaike's Information Criterion (AIC), ΔAIC , and changes in model deviance (Dev).

Model	AIC	ΔAIC	K	Dev
Total ducks	4,426.71	−16.62	17	4,392.71
Random 5	4,439.95	−3.38	17	4,405.95
Random 4	4,442.20	−1.13	17	4,408.19
Vegetative cover	4,442.65	−0.68	17	4,408.65
Random 8	4,442.81	−0.52	17	4,408.80
Random 1	4,442.90	−0.43	17	4,408.90
Base model	4,443.34	0	16	4,411.33
Not Sunday, Monday, or Wednesday	4,445.09	1.75	17	4,411.08
Cover type	4,445.25	1.92	17	4,411.25
Last duck seen a mallard	4,445.33	1.99	17	4,411.32



> dredge (no)

PPPs; BCIs; CIs; p-values; DIC; AIC; BIC; AICc; WAIC; Bayesian p-values; deviance; egads!

Two primary goals today

- i. Review model selection (Script 7c)**
- 1. Examine how Bayesian p-values work (Script 7d)**

i. One last (simulation-based) model-selection example...

What is ‘likelihood’?: It’s exactly what it sounds like...

$$l(y_i | \boldsymbol{\beta}) = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln(\sigma^2) - \frac{1}{2\sigma^2} \left(\sum_{i=1}^n (y_i - \mu_i)^2 \right)$$

We use likelihood functions to estimate the **most likely parameter values** (i.e., maximum likelihood estimation)

We use likelihood functions to **compare models** (i.e., model selection)

There are different likelihood functions for different error distributions

Poisson regression log-likelihood

$$l(y_i|\boldsymbol{\beta}) = \sum_{i=1}^n y_i \times \ln(\mu_i) - \sum_{i=1}^n \mu_i - \sum_{i=1}^n \ln(y_i!)$$

Linear regression log-likelihood

$$l(y_i|\boldsymbol{\beta}) = -\frac{n}{2}\ln(2\pi) - \frac{n}{2}\ln(\sigma^2) - \frac{1}{2\sigma^2} \left(\sum_{i=1}^n (y_i - \mu_i)^2 \right)$$

We use likelihood functions and a parameter penalty to rank models

AIC (general equation)

This is the log-likelihood (we work on the log-scale to prevent extreme values)

$$AIC = 2k - 2 * \ln(\hat{L})$$

k is the number of parameters (this penalizes complex models)

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AICc

$$AICc = AIC + \frac{2k^2 + 2k}{n - k - 1}$$

What is DIC?

Deviance = -2 x log-likelihood

Effective number of parameters

$$\text{DIC} = \boxed{\bar{D}} + \boxed{pD}$$

$$\text{AICc} = \boxed{-2\log L} + \boxed{2k + \frac{2k^2 + 2k}{n - k - 1}}$$

Deviance = -2 x log-likelihood

Parameter penalty

They're very similar!

How do we get the log-likelihood?

$$\text{DIC} = \bar{D} + pD$$

$$pD = \left(\frac{\sigma_D^2}{2} \right)$$

We do this for every iteration

$$\log L = -\frac{n}{2} \times \ln(2\pi) - \frac{n}{2} \ln(\sigma^2) - \frac{1}{2\sigma^2} \times \sum (\mathbf{y} - \beta \mathbf{x})^2$$

$$D = -2 \times \log L$$

Well, JAGS (and Stan and Nimble) calculates it for us!

What is WAIC?

WAIC is very similar to AICc and DIC

We do this for every point at every iteration

$$\log L = -\frac{1}{2} \times \ln(2\pi) - \frac{1}{2} \ln(\sigma^2) - \frac{1}{2\sigma^2} \times (y - \beta x)^2$$

WAIC is very similar to AICc and DIC

- 1) We then take the mean of the likelihood (not log-likelihood) for each point across all iterations
- 2) We take the log of those means
- 3) And then we sum across points to get log pointwise predictive density (lppd)

WAIC is very similar to AICc and DIC

- 1) To get the effective number of parameters, we sum the variances of the log-likelihoods for each point

$$\text{WAIC} = -2 \times (\text{lppd} - p_{\text{WAIC}})$$

$$\text{DIC} = \bar{D} + pD$$

$$\text{AICc} = -2\log L + 2k + \frac{2k^2 + 2k}{n - k - 1}$$

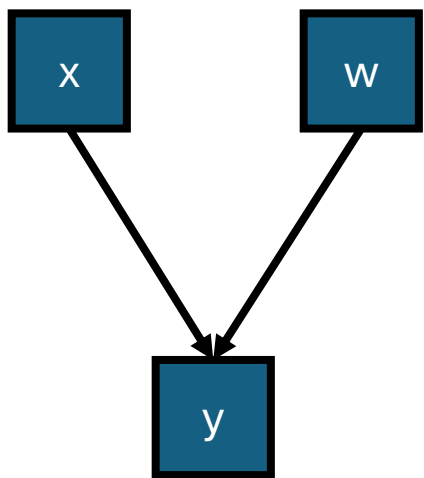
Let's work through an example

The data-generating model

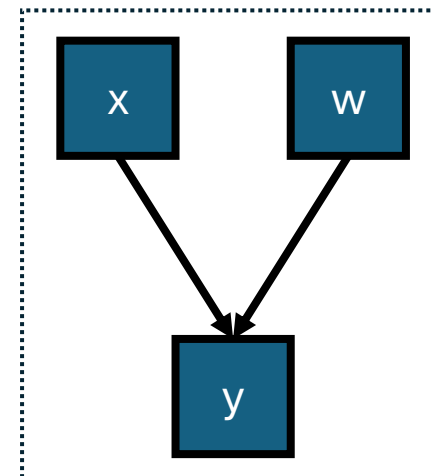
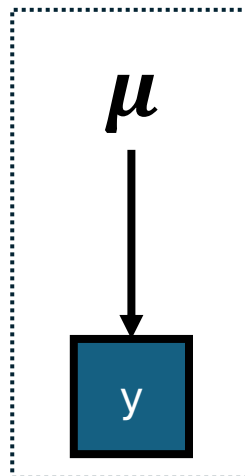
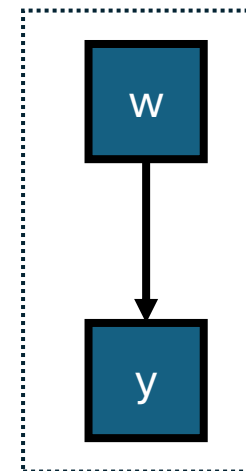
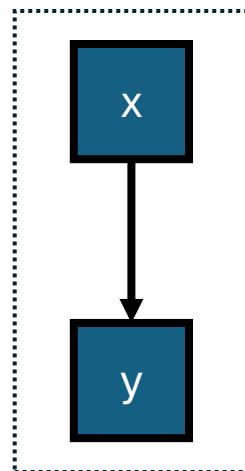
$$\mathbf{x} \sim \text{Normal}(0, \sigma = 1)$$

$$\mathbf{w} \sim \text{Normal}(0, \sigma = 0.05)$$

$$\mathbf{y} \sim \text{Normal}(\mathbf{x} + \mathbf{w}, \sigma^2 = 1)$$

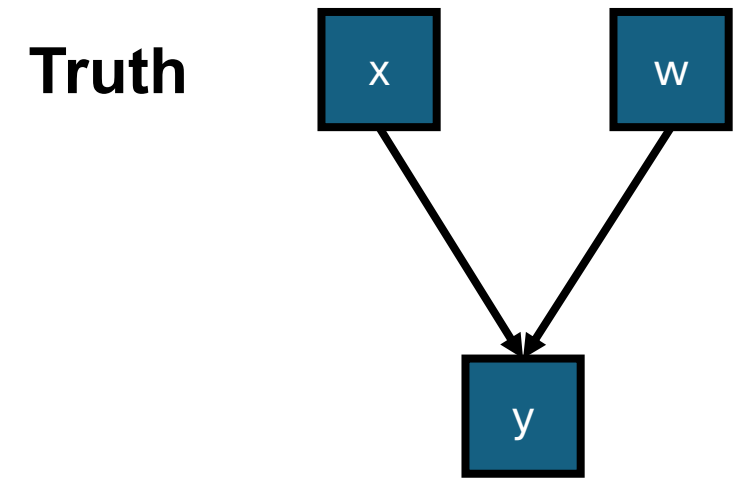
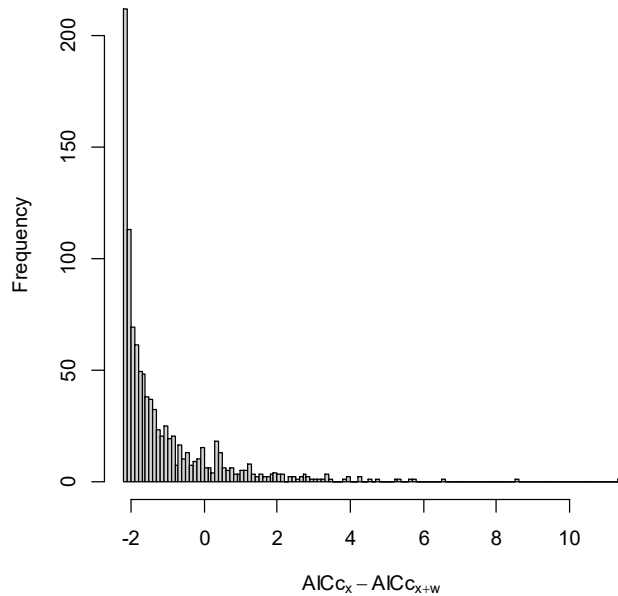


Competing models

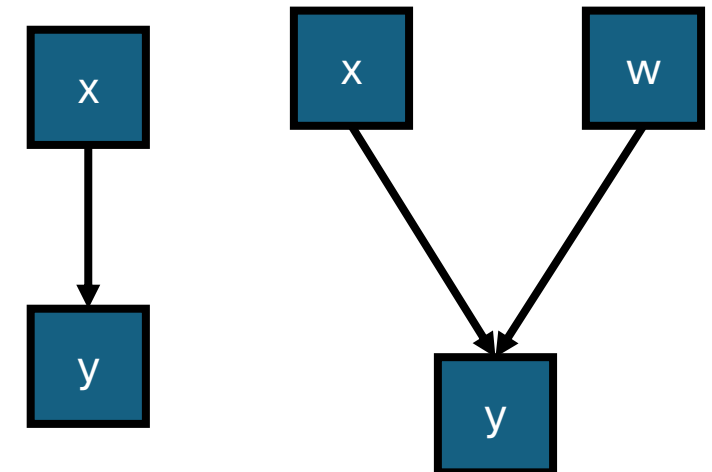


Model selection (AICc only)

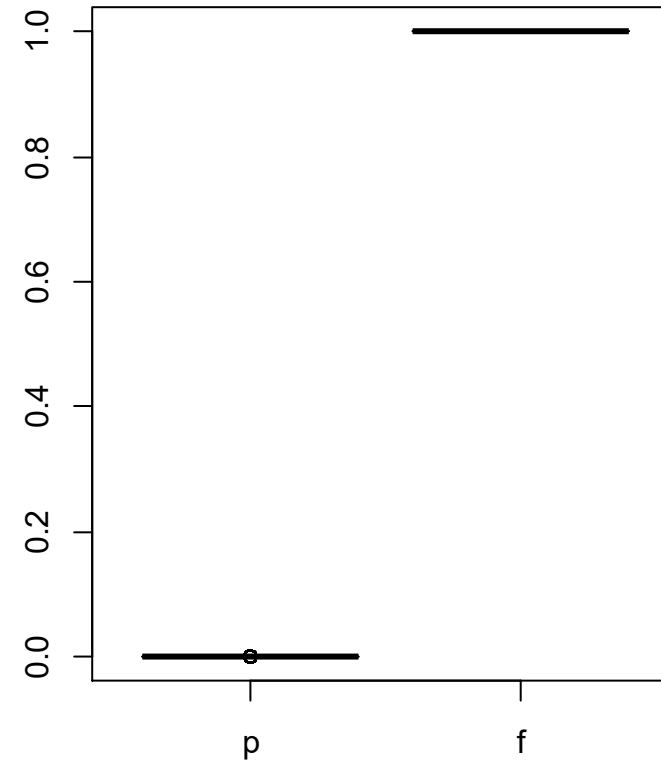
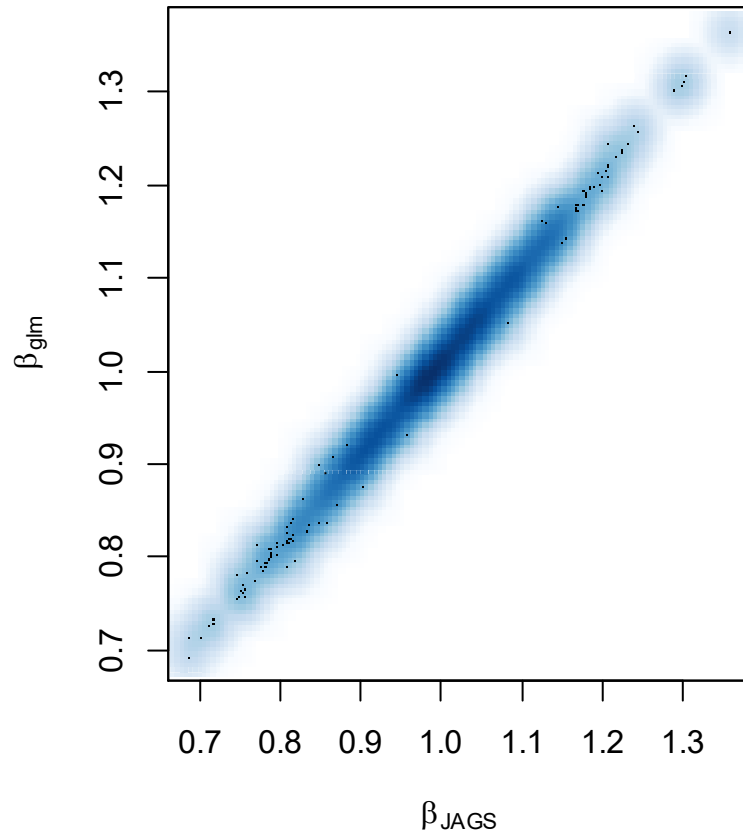
- The model that only included x was the top model $\sim 85\%$ of the time.
- The data-generating model was the top model 15% of the time. Why?



Competing models

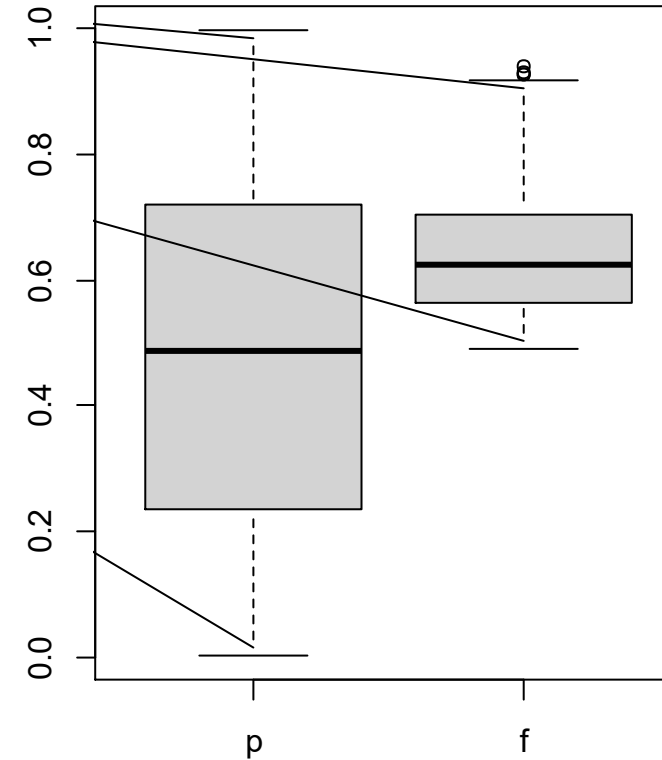
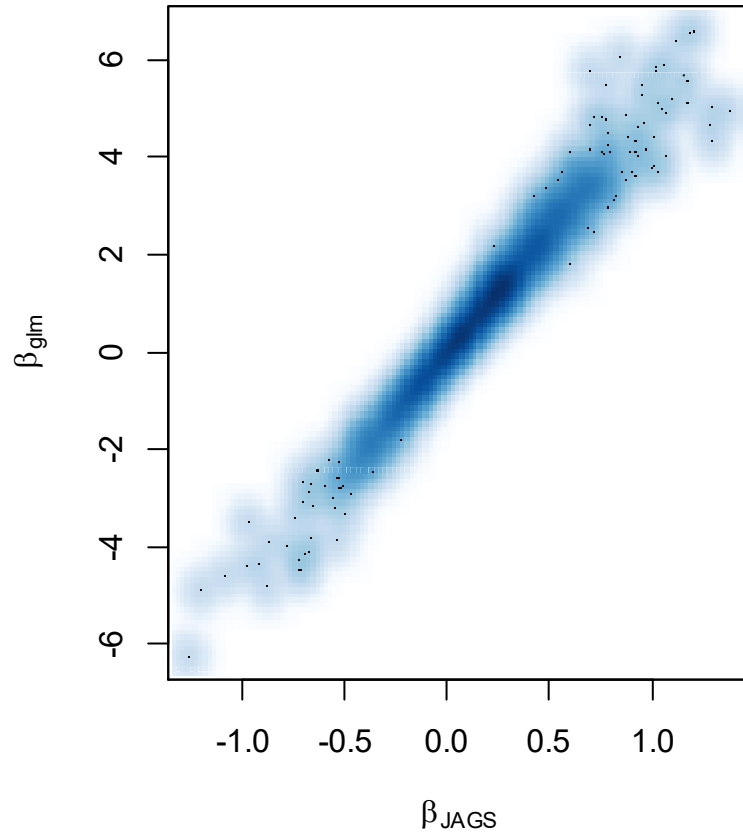


Estimates of the effect of x from the data generating model



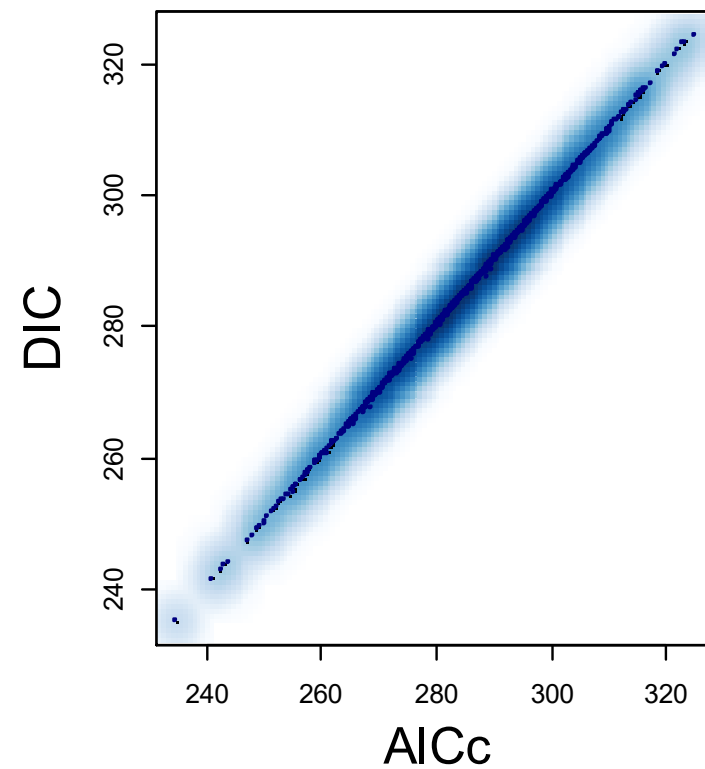
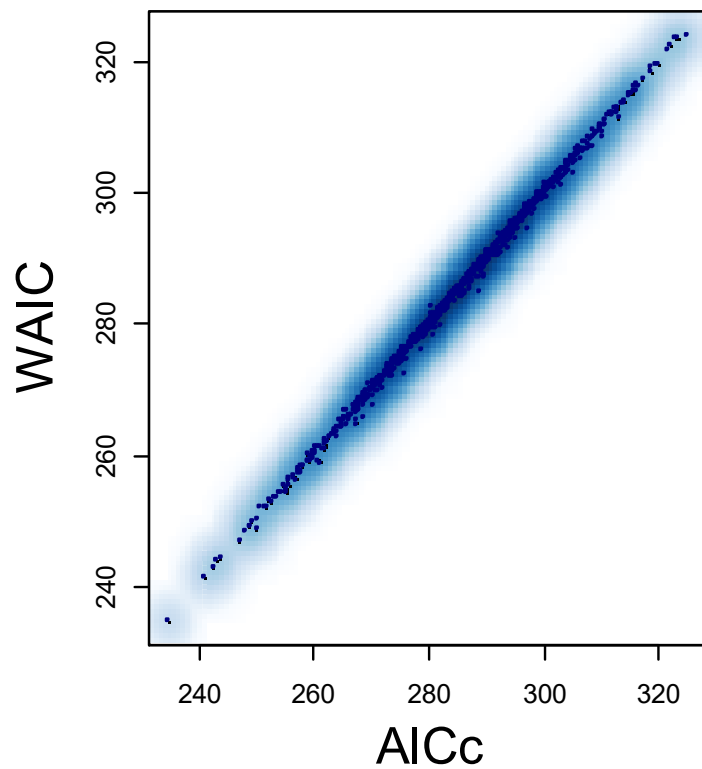
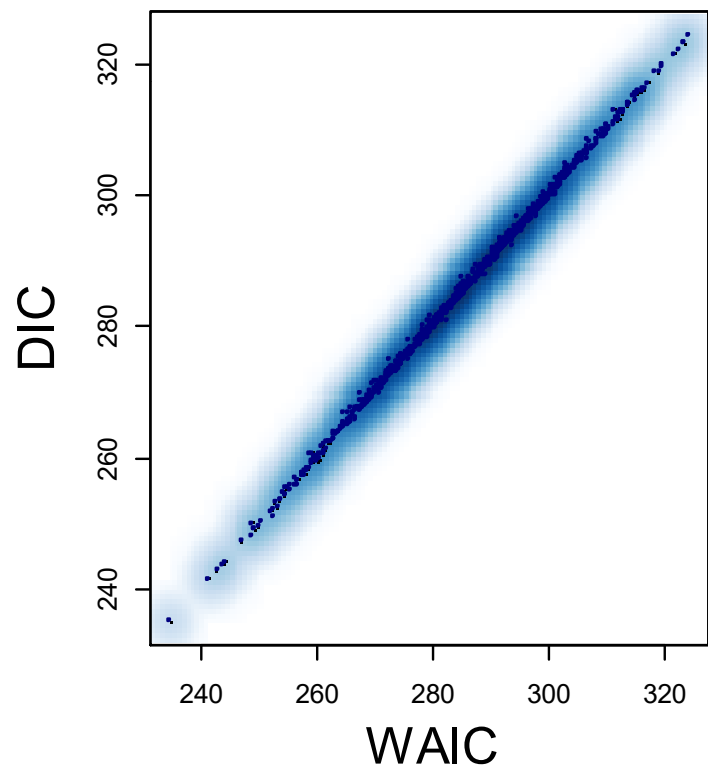
All simulations: $p < 0.01$ & $f > 0.99$

Estimates of the effect of w from the data generating model

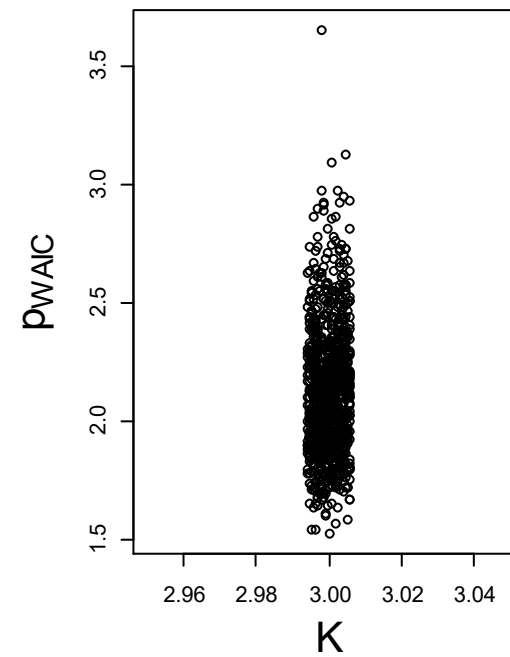
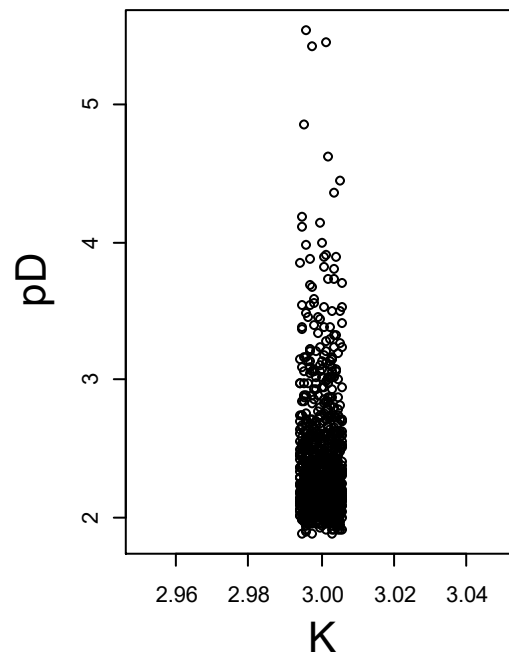
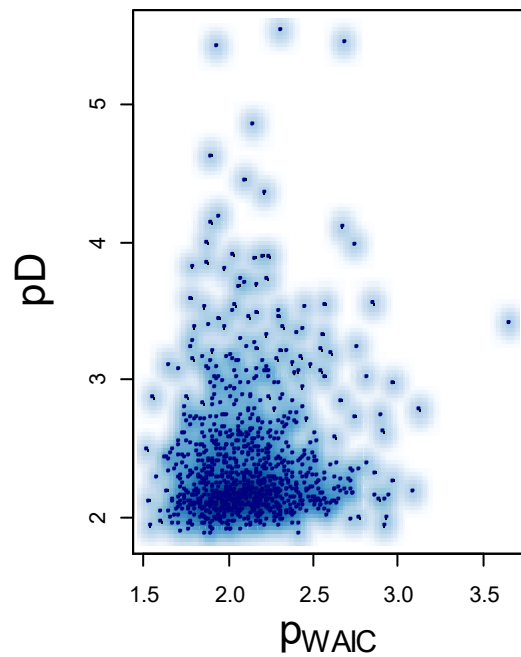


$p < 0.05$ in 59 of 1000 simulations, f was > 0.95 zero times out of 1000

Estimates of different information criteria



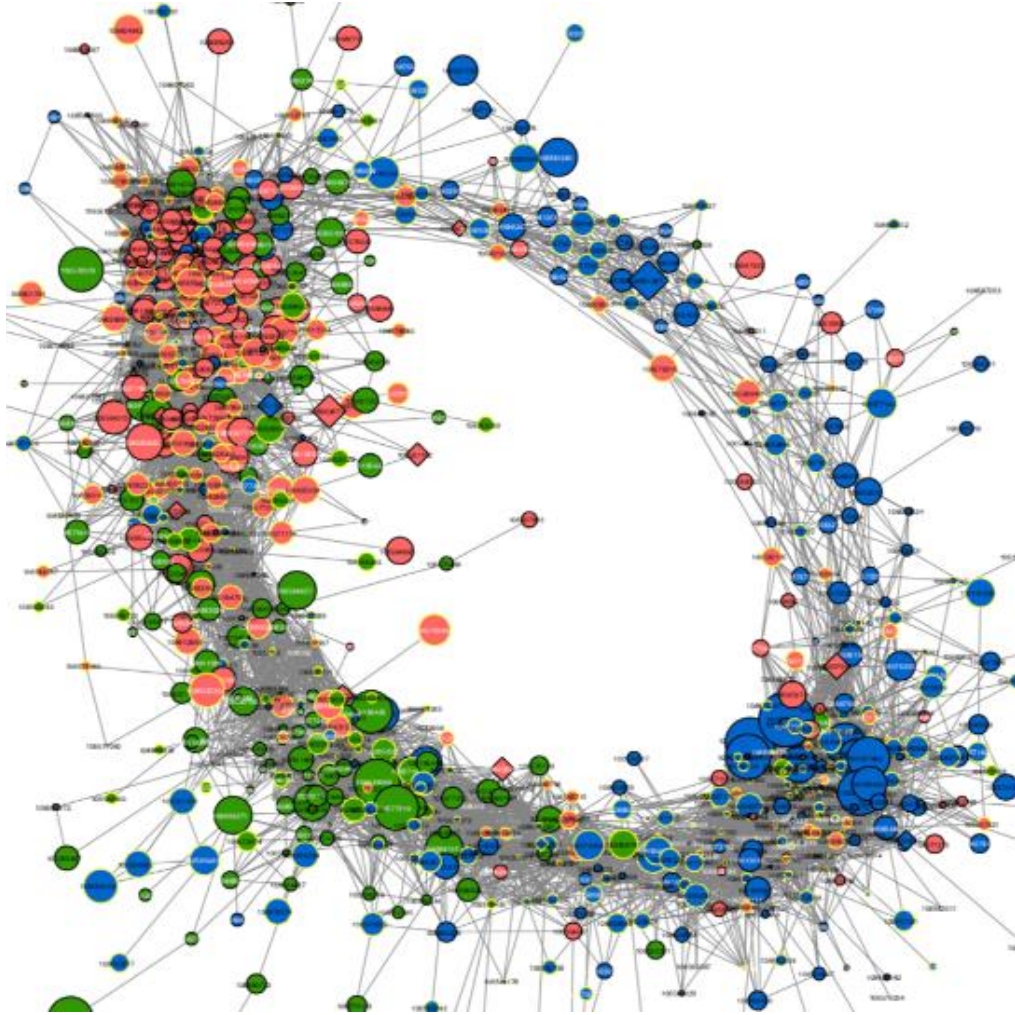
Estimates of the ‘number of parameters’



What are the key take-homes...

i.e., what's the point of all of this?

This is [a gross oversimplification of] reality

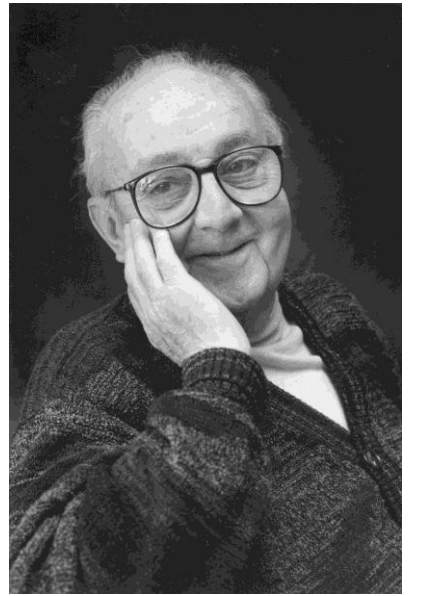


Everything that we know about the world is a model. Every word and language is a model. All maps and statistics, books and databases, equations and computer programs are models. So are the ways I picture the world in my head – my mental models. None of these is or ever will be the real world.

- Donella Meadows



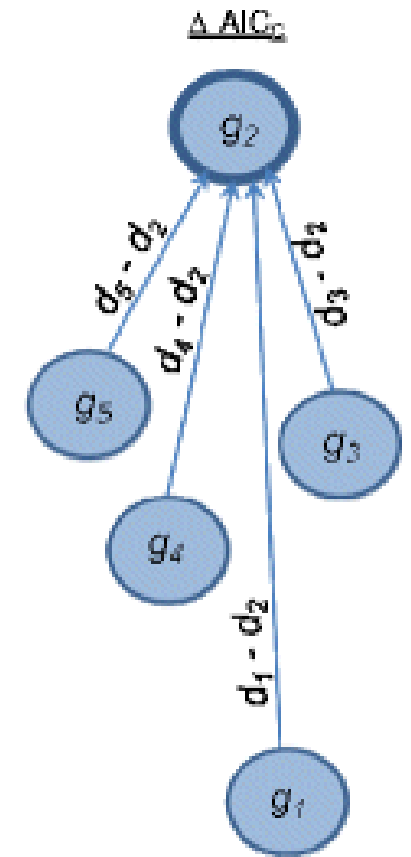
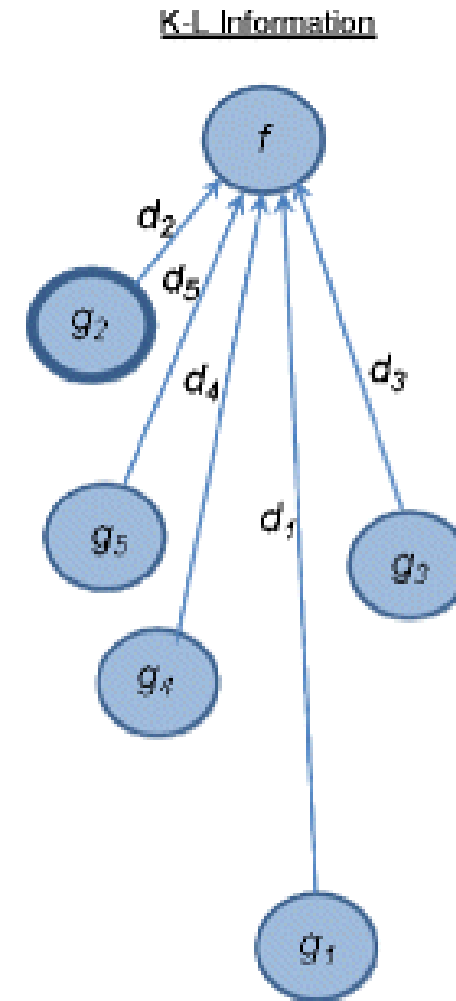
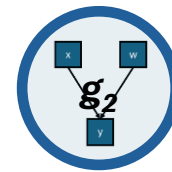
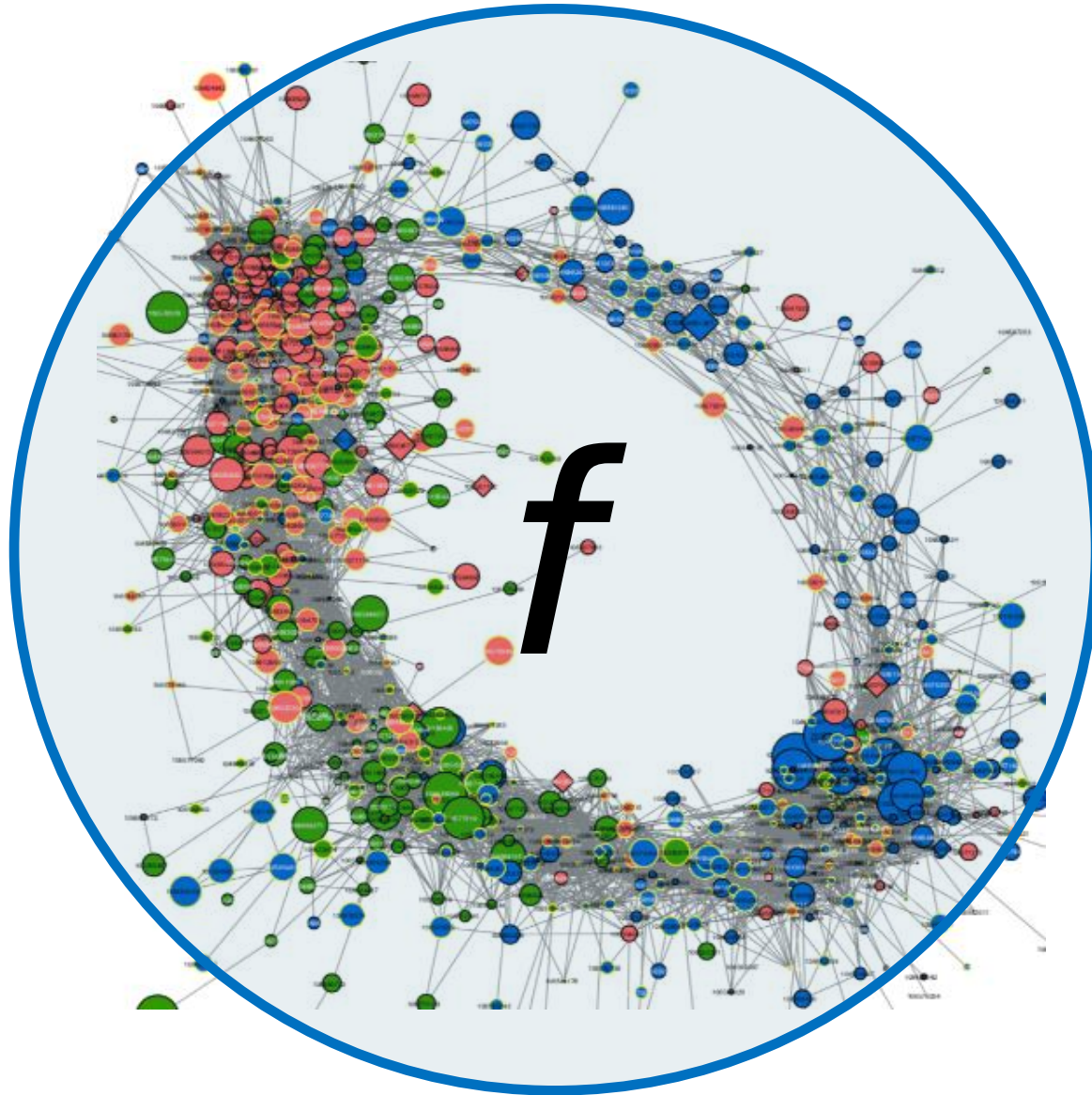
All models are wrong, but some are useful - **George EP Box**



No models are right, and most are useless - **Todd Arnold**



d_1 through d_5 are “pretty big”, δAIC is the difference b/w d_1 through d_5



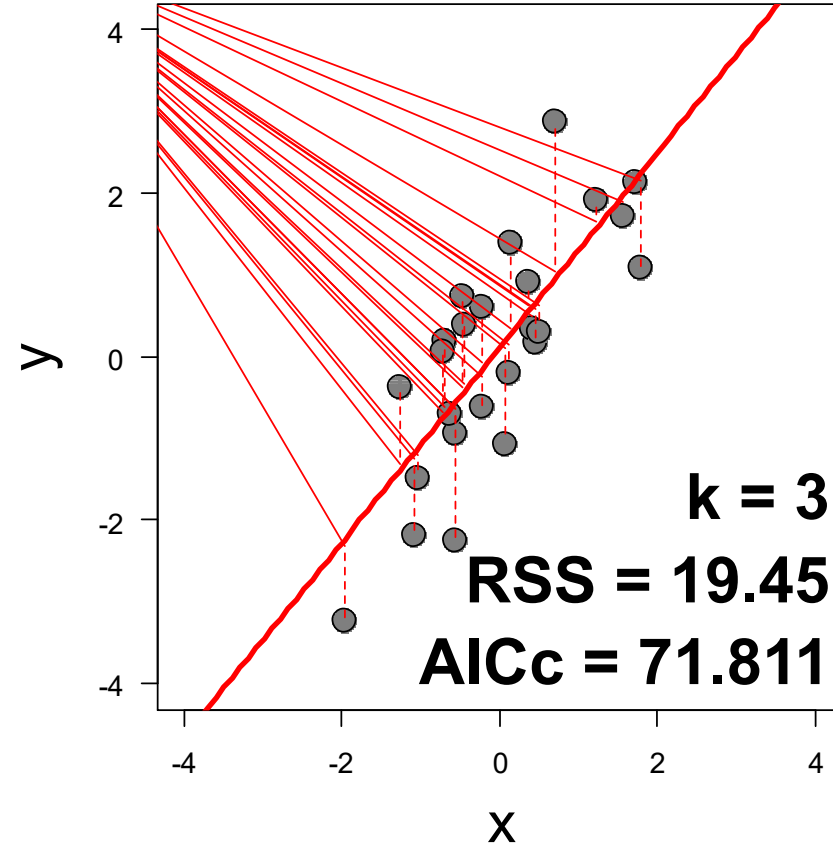
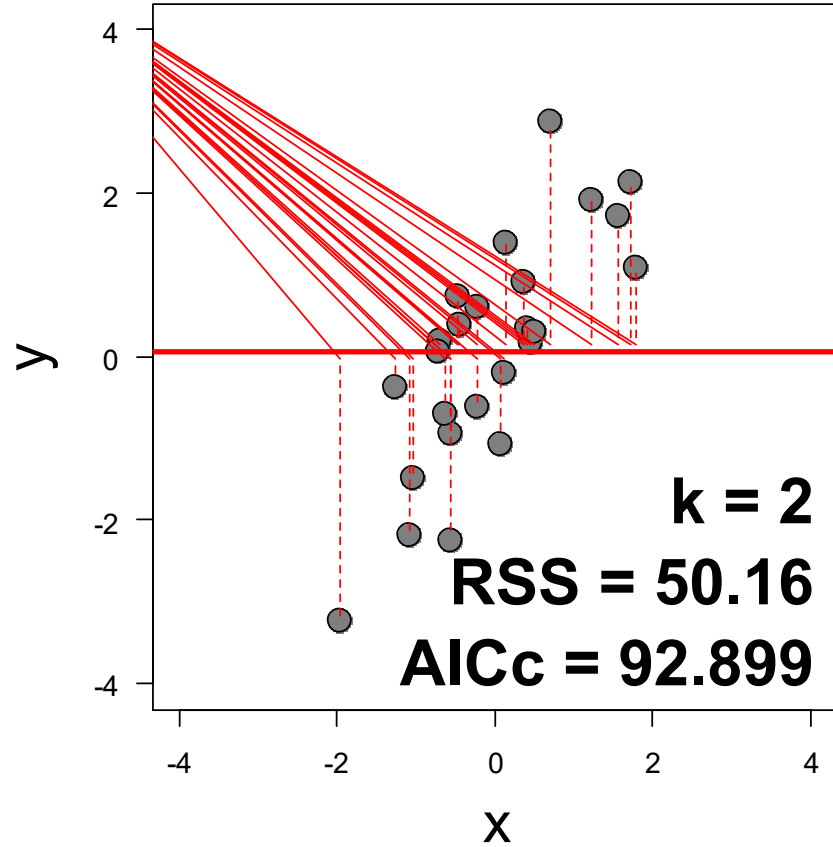
A model selection table tells us whether RSS changed enough to offset δK ...

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$$AICc = 2k + n \times \left(\ln \left(2 \times \pi \times \frac{RSS}{n} \right) + 1 \right) + \frac{2k^2 + 2k}{n - k - 1}$$

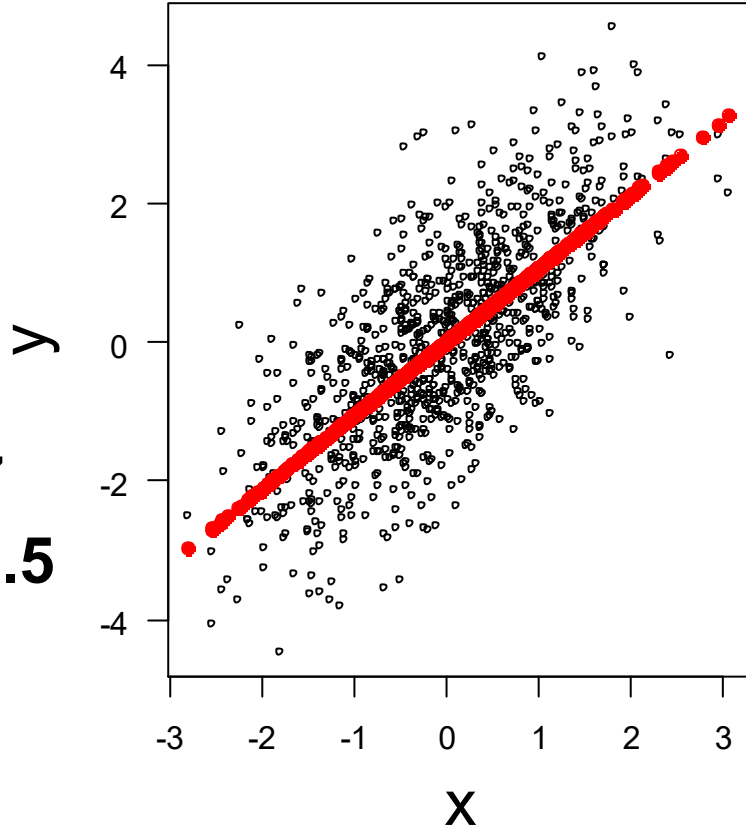
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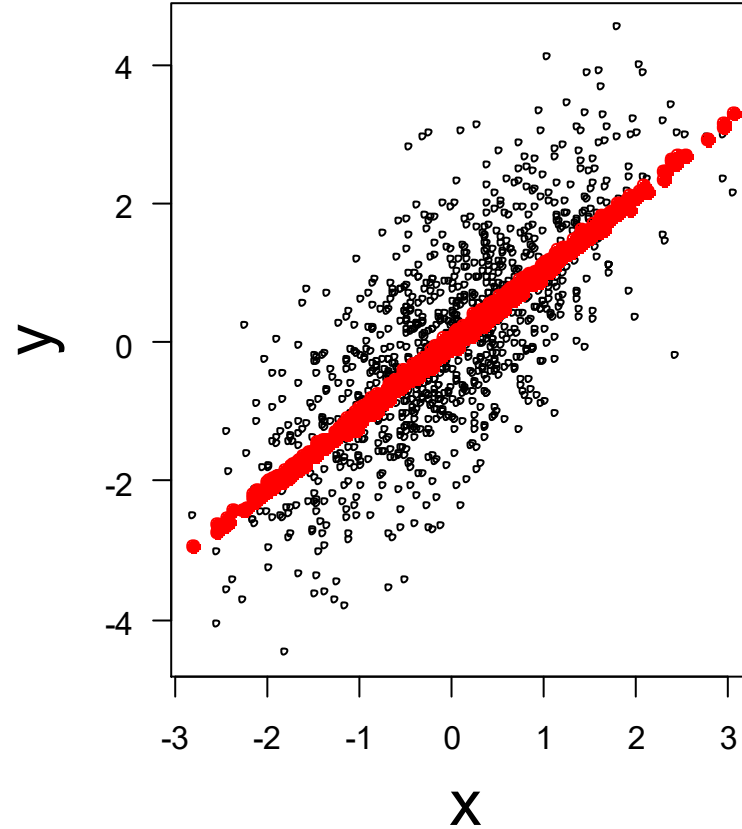
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A model selection table tells us whether RSS changed enough to offset δK ...

k = 2
RSS = 990.7
AICc = 2832.5



k = 3
RSS = 987.9
AICc = 2831.8



$$AICc = 2k + n \times \left(\ln \left(2 \times \pi \times \frac{RSS}{n} \right) + 1 \right) + \frac{2k^2 + 2k}{n - k - 1}$$

I don't expect you to memorize this stuff, but...

$$l(y_i | \boldsymbol{\beta}) = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln(\sigma^2) - \frac{1}{2\sigma^2} \left(\sum_{i=1}^n (y_i - \mu_i)^2 \right)$$

I want you to know that a difference of Δ AIC means that...

$$\text{AICc} = 2k + n \times \left(\ln \left(2 \times \pi \times \frac{\text{RSS}}{n} \right) + 1 \right) + \frac{2k^2 + 2k}{n - k - 1}$$

The predictions got close(r) enough to the observations to offset the k penalty

For simple models, there will be no meaningful differences in model selection among DIC, WAIC, AICc, BIC, etc.

1. **A first stab at goodness-of-fit testing:**
what are Bayesian p -values?

Why do I have to know goodness-of-fit if I already have my best model?

It's important.

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What is goodness-of-fit?

Yes. This should have been your first question.

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No.

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Not exactly.

What is goodness-of-fit?

Yes. This should have been your first question.

Where can I find more information about goodness-of-fit, particularly for extremely complex models?

Information will mostly be conveyed by statisticians who know so much more about notation than you that you won't be able to understand what they're saying

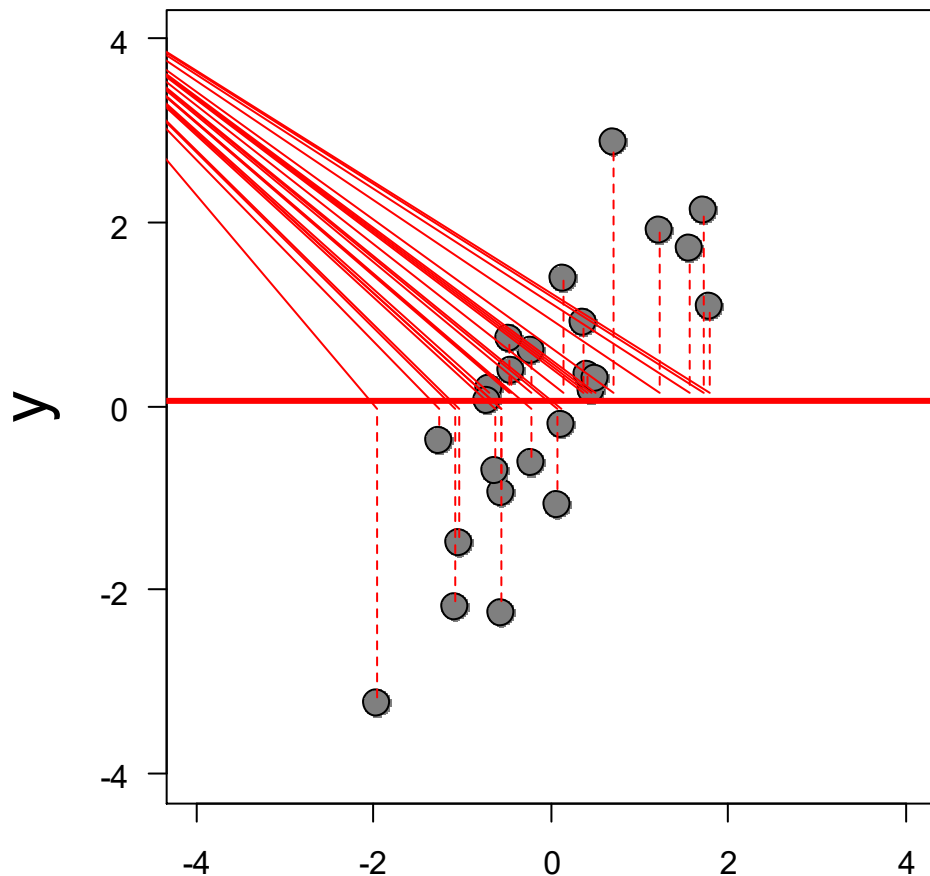
What is goodness of fit?

- Most [all?] goodness-of-fit metrics rely on comparing your actual data to data that are simulated from the model
- We'll simulate new data (y') from our parameter estimates at every iteration

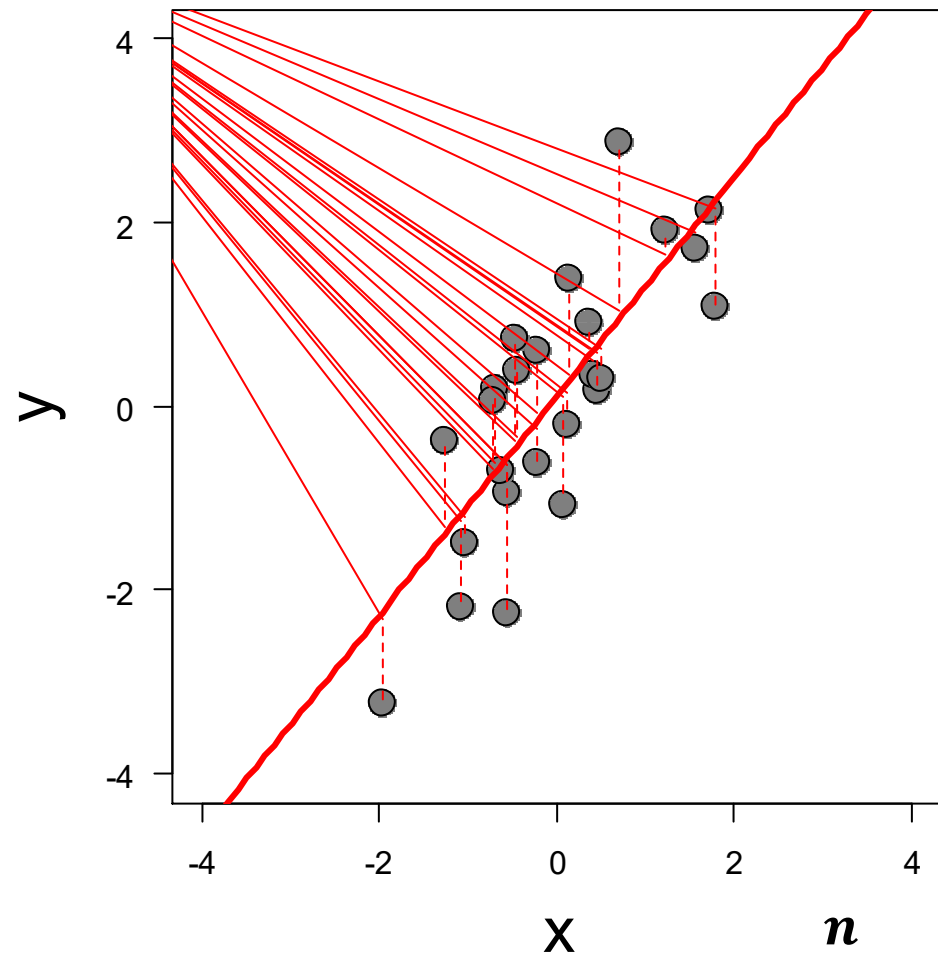
$$\mathbf{y} \sim \text{Normal}(\beta \mathbf{x}, \sigma^2)$$

$$\mathbf{y}' \sim \text{Normal}(\beta \mathbf{x}, \sigma^2)$$

Remember residual sums of squares?



$$50.156 = \sum_{i=1}^n (y_i - \eta_i)^2$$



$$19.448 = \sum_{i=1}^n (y_i - \eta_i)^2$$

Much like likelihoods, we can use different test statistics for different error distributions, i.e., you might calculate p-values in different ways for different model types

What are Bayesian p-values?

- We'll simulate new data (y') from our parameter estimates at every iteration

$$\mathbf{y} \sim \text{Normal}(\beta \mathbf{x}, \sigma^2)$$

$$\mathbf{y}' \sim \text{Normal}(\beta \mathbf{x}, \sigma^2)$$

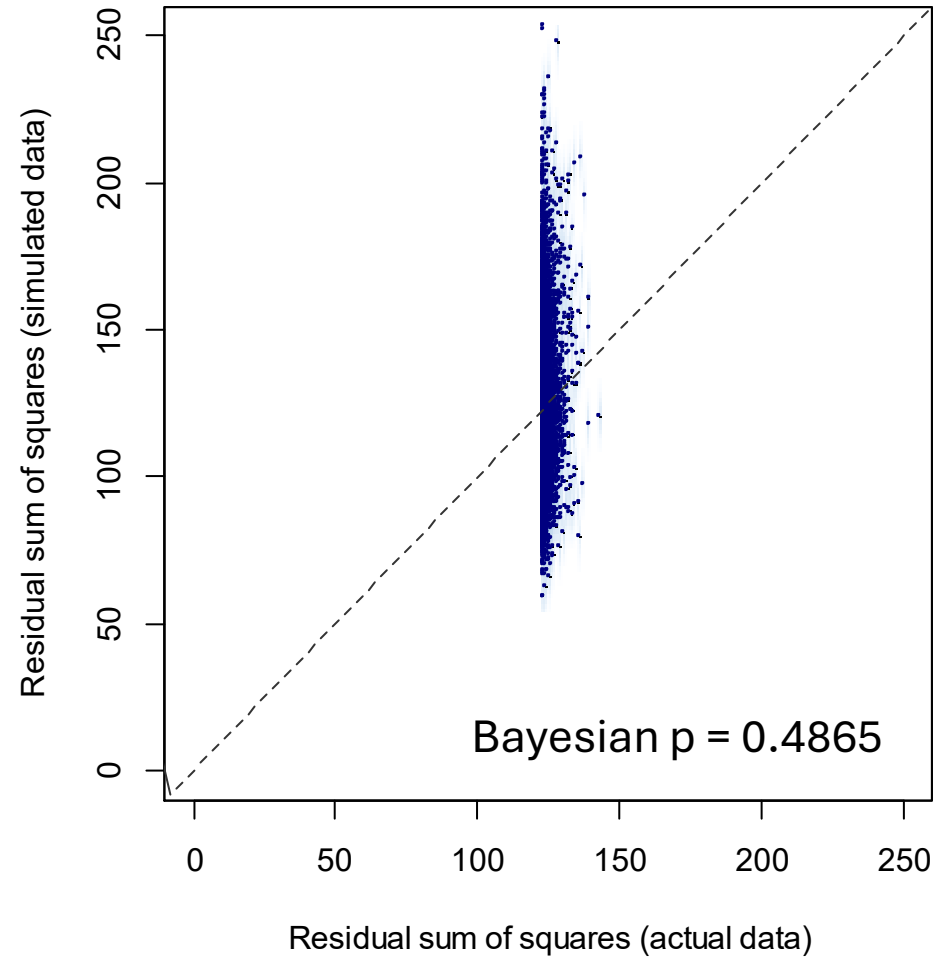
- We'll calculate RSS for the new data (and the real data, note we often use other discrepancy statistics for other model types)
- We'll compare the RSS from simulated and real data. The p-value is the proportion of times that the RSS from the real data is greater than the RSS from the simulated data

What are acceptable Bayesian p-values?

- A fitting model has a Bayesian p-value near 0.5, and values close to 0 or close to 1 suggest doubtful fit of the model (Kéry [2010] *Academic Press*)
- Values less than 0.1 or greater than 0.9 suggest our model does not fit the data well (Hobbs and Hooten [2015] *Princeton*)

What am I about to say?

What are Bayesian p-values?



Why is the range of RSS smaller for the actual data?

Ok, let's break some stuff again

Why not? It'll be fun

We're going to simulate clutch size as a function of body size

$$n = 200$$

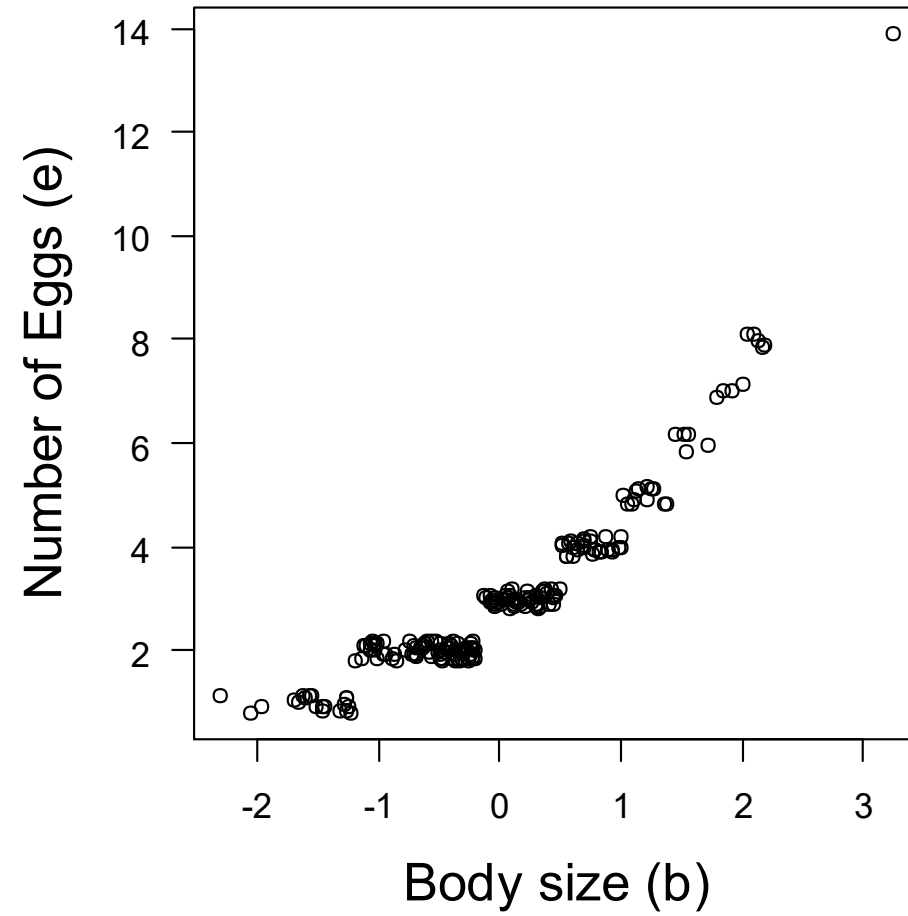
$$\mathbf{b} \sim \text{Normal}(0,1)$$

$$\psi = e^{\alpha + \beta \times b}$$

$$\mathbf{e} \sim \text{round}(\psi)$$

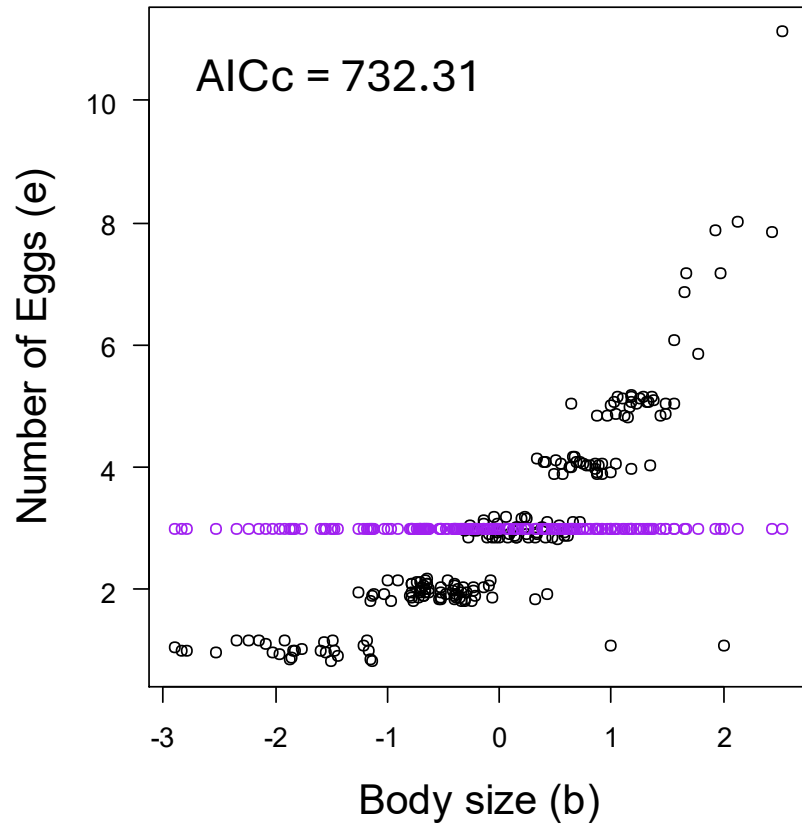
$$\alpha = 1$$

$$\beta = 0.5$$

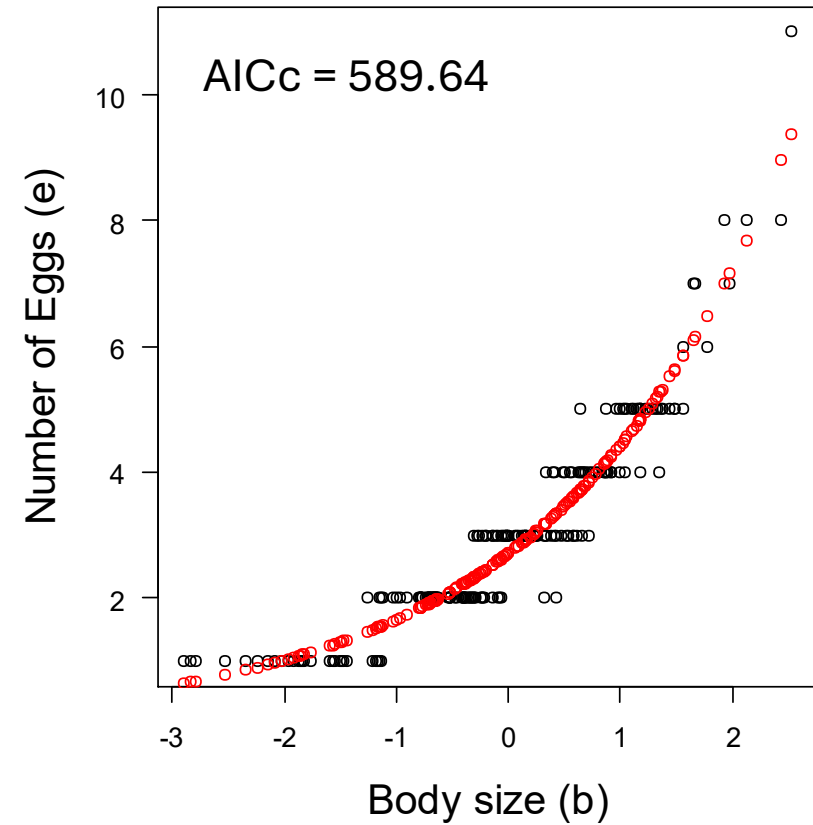


Residuals and AICc...

```
glm(y ~ 1, family = 'poisson')
```



```
glm(y ~ b, family = 'poisson')
```



$$\bar{e} = 3.02, \sigma_e^2 = 2.91$$

Deviance (D) Information Criterion (DIC; run the same models in JAGS)

Model 0

$$\psi = e^{\alpha}$$

$$e \sim \text{Poisson}(\psi)$$

$$\alpha \sim \text{Normal}(1,1)$$

$$\text{DIC} = \bar{D} + 2pD$$

$$pD = \frac{\sigma_D^2}{2}$$

Model 1

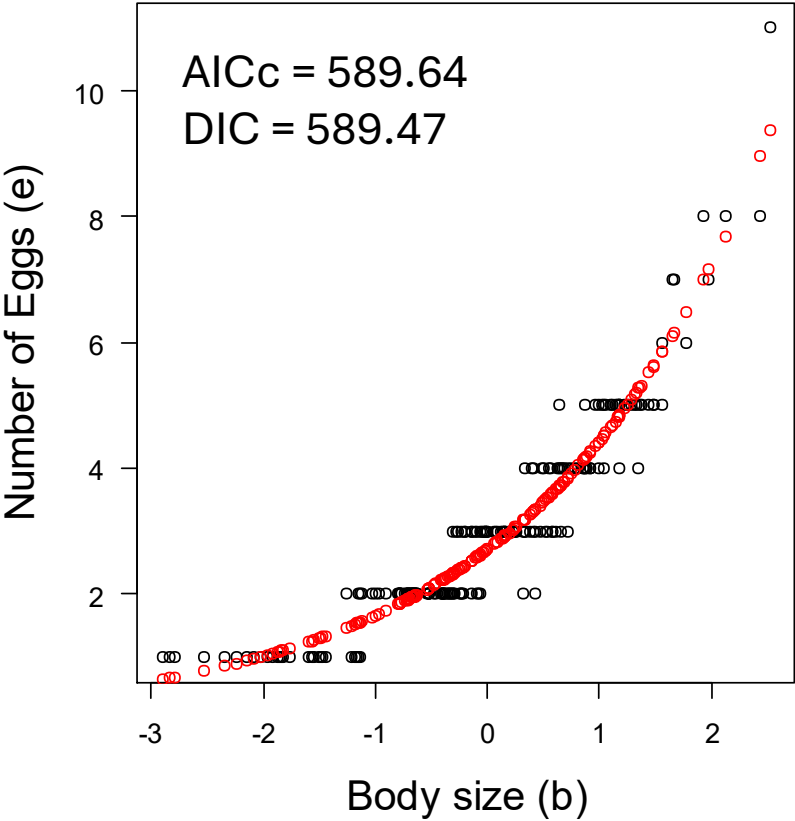
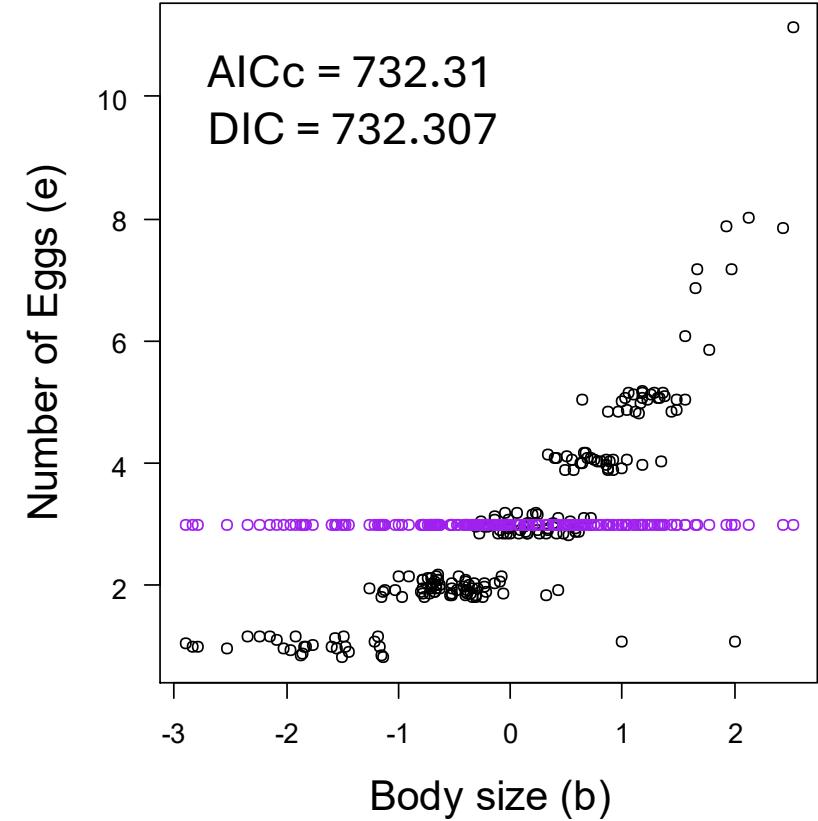
$$\psi = e^{\alpha + \beta \times b}$$

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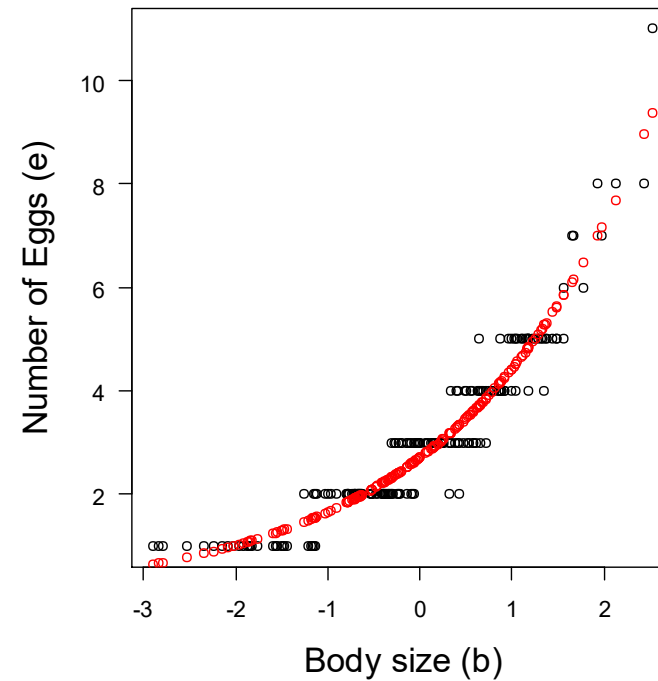
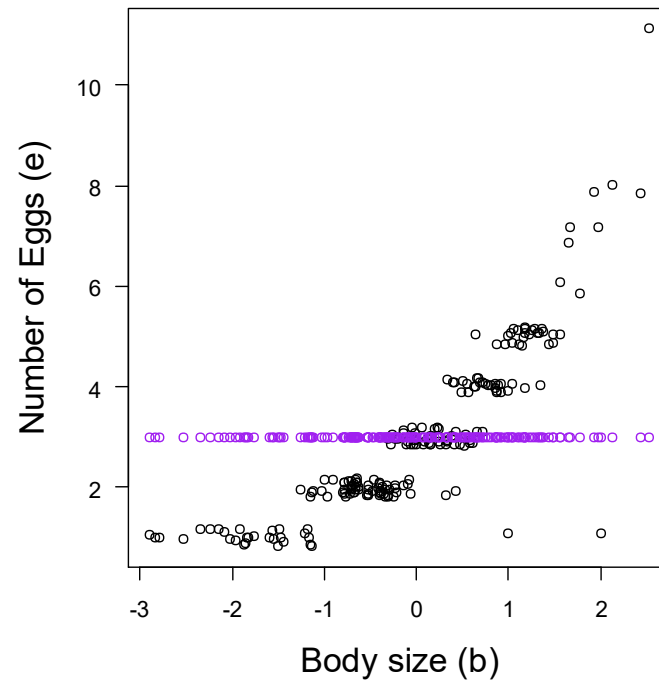
$$\alpha \sim \text{Normal}(1,1)$$

$$\beta \sim \text{Normal}(0,1)$$

DIC



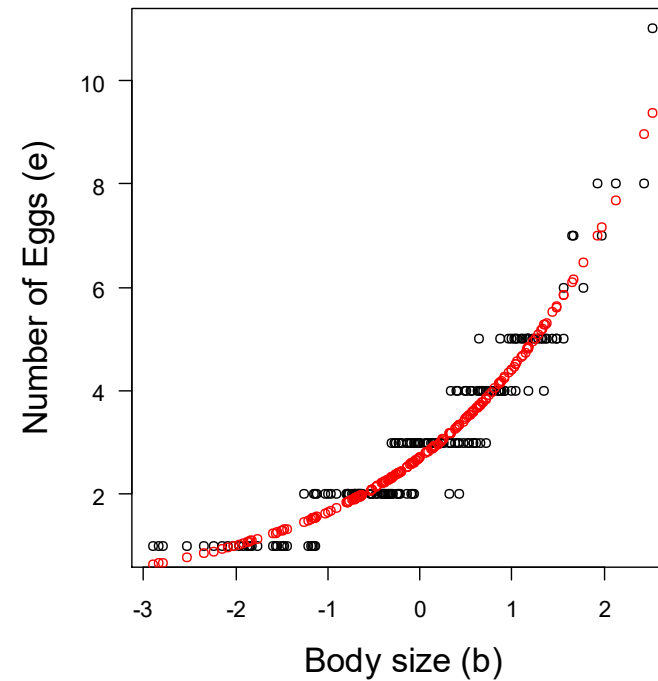
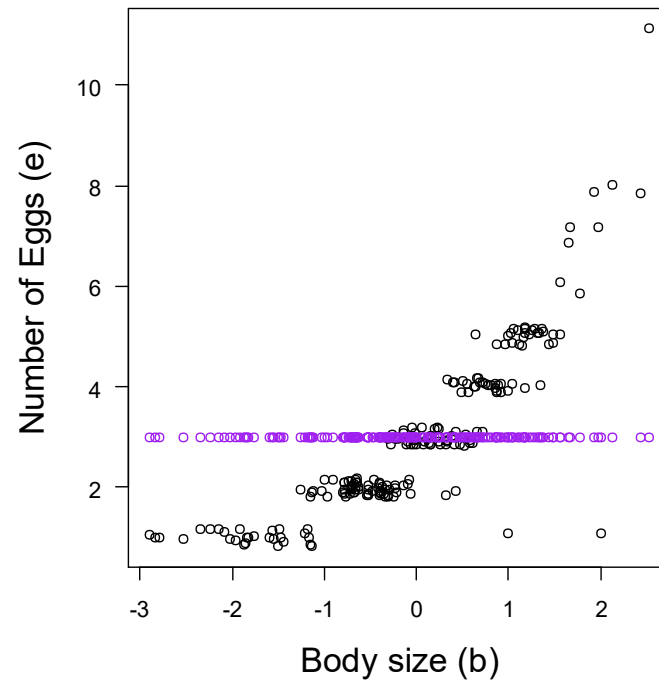
Bayesian p-values (RSS for this Poisson regression)



1. Calculate (Pearson's) residuals at each iteration

$$\varepsilon = \frac{e - e^{\beta_0 + \beta_1 \times b}}{\sigma_e}$$

Bayesian p-values (RSS for this Poisson regression)

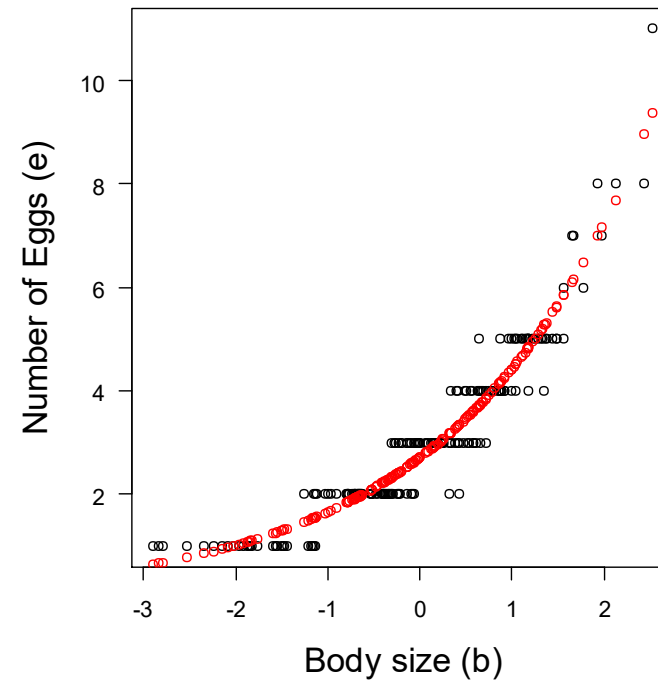
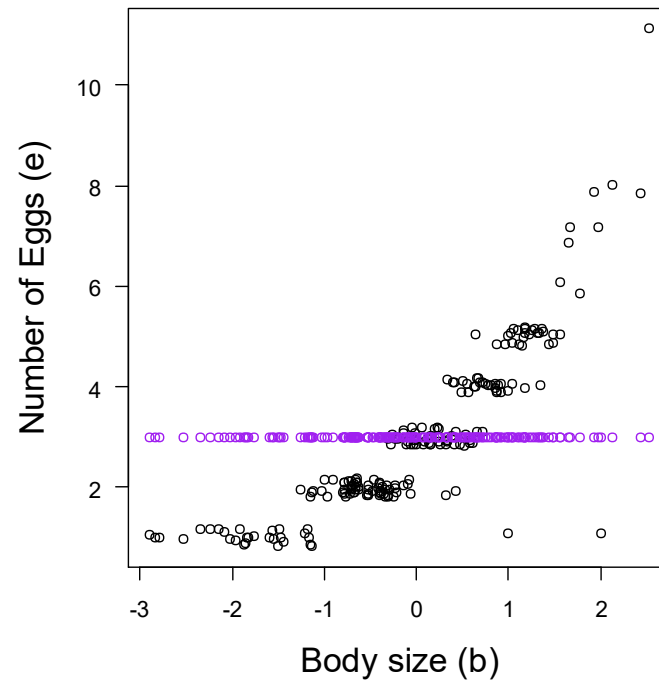


1. Calculate (Pearson's) residuals at each iteration
2. Generate 'new' data

$$e \sim \text{Poisson}(\psi)$$

$$e' \sim \text{Poisson}(\psi)$$

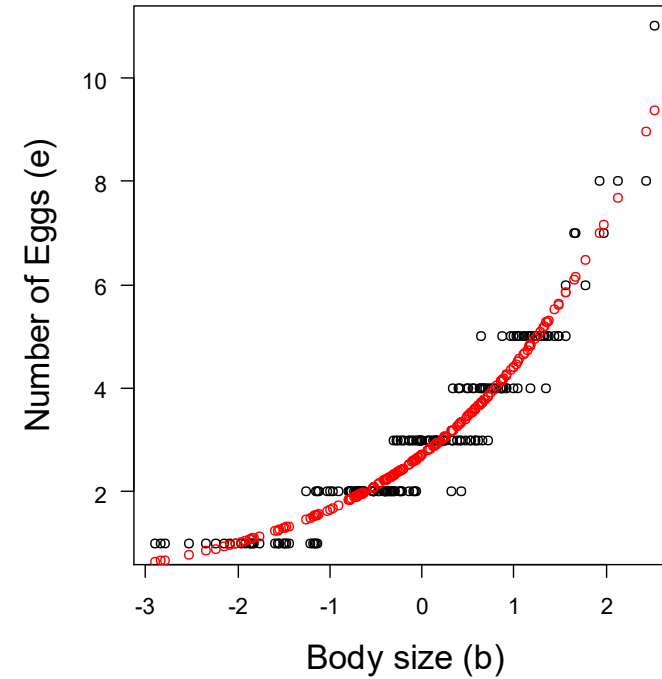
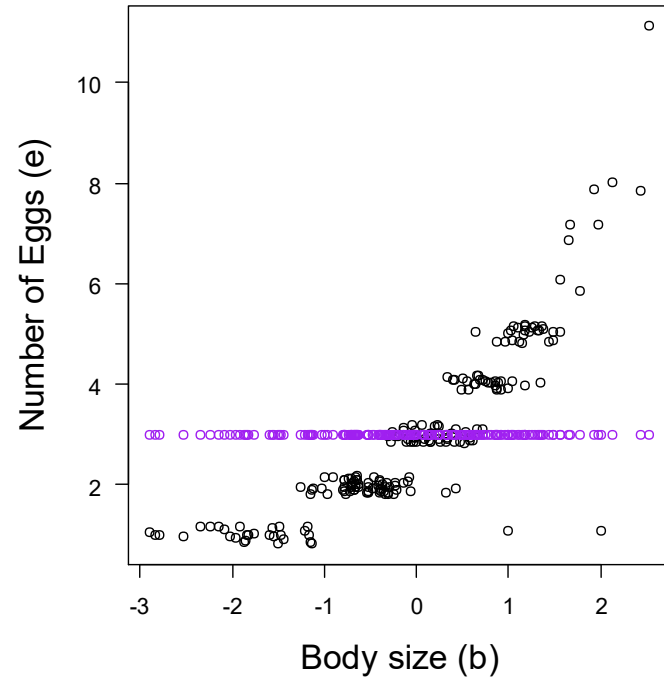
Bayesian p-values (RSS for this Poisson regression)



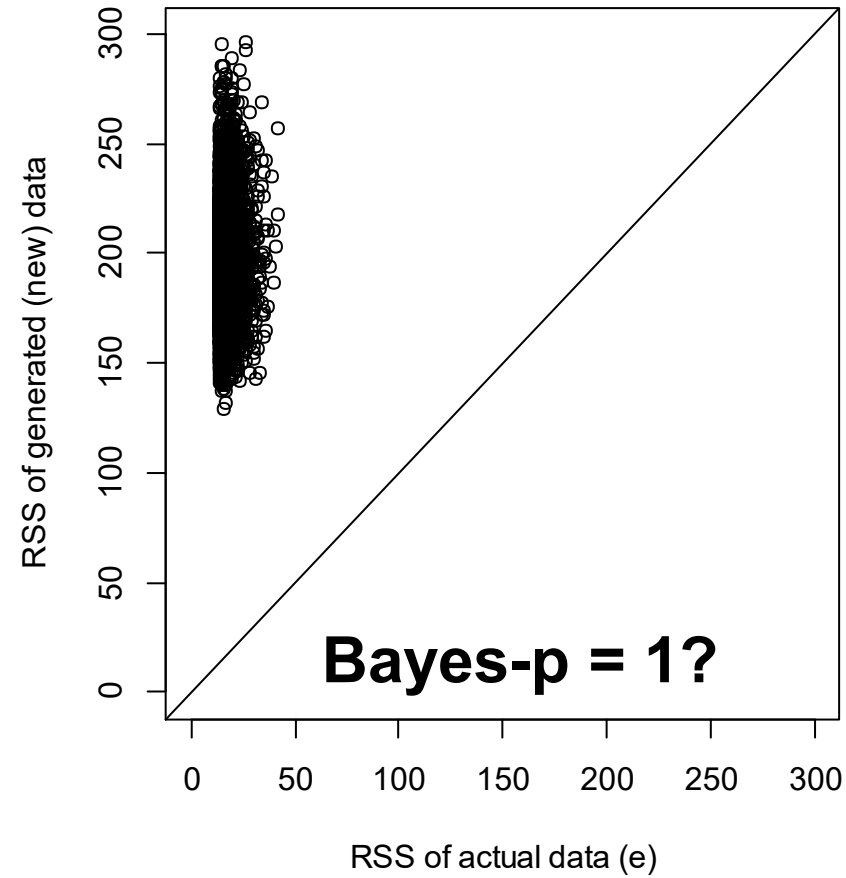
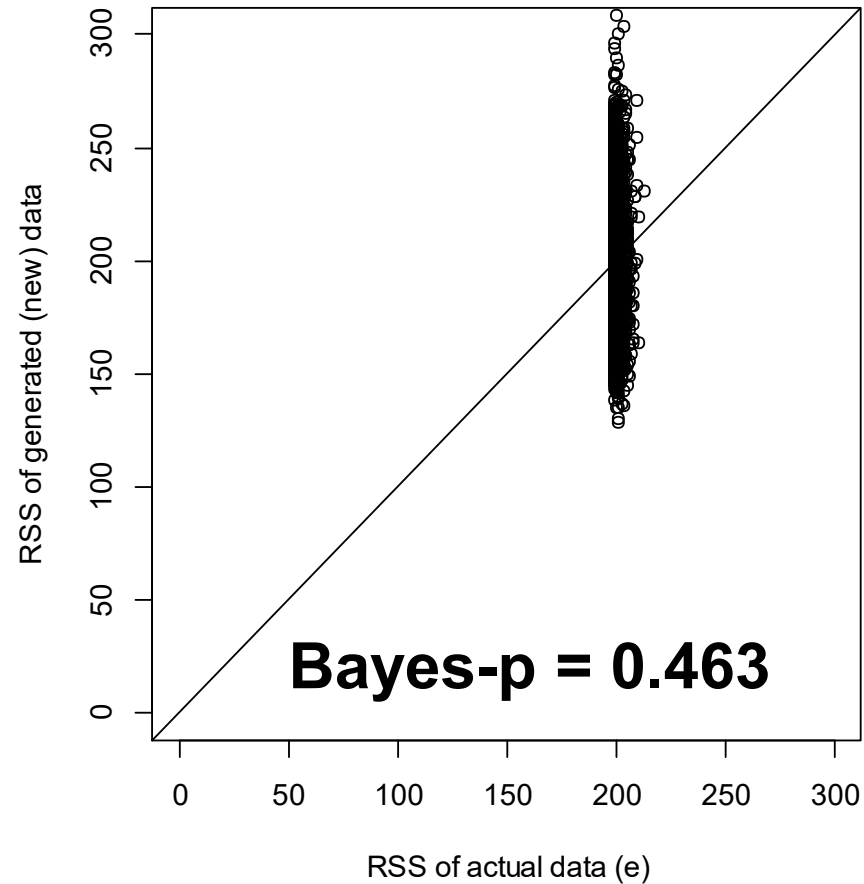
1. Calculate (Pearson's) residuals at each iteration
2. Generate 'new' data
3. Calculate (Pearson's) residuals at each iteration for new data

$$\varepsilon' = \frac{e' - e\beta_0 + \beta_1 \times b}{\sigma_{e'}}$$

Which model will fit the data better (have a Bayes p closer to 0.5)?

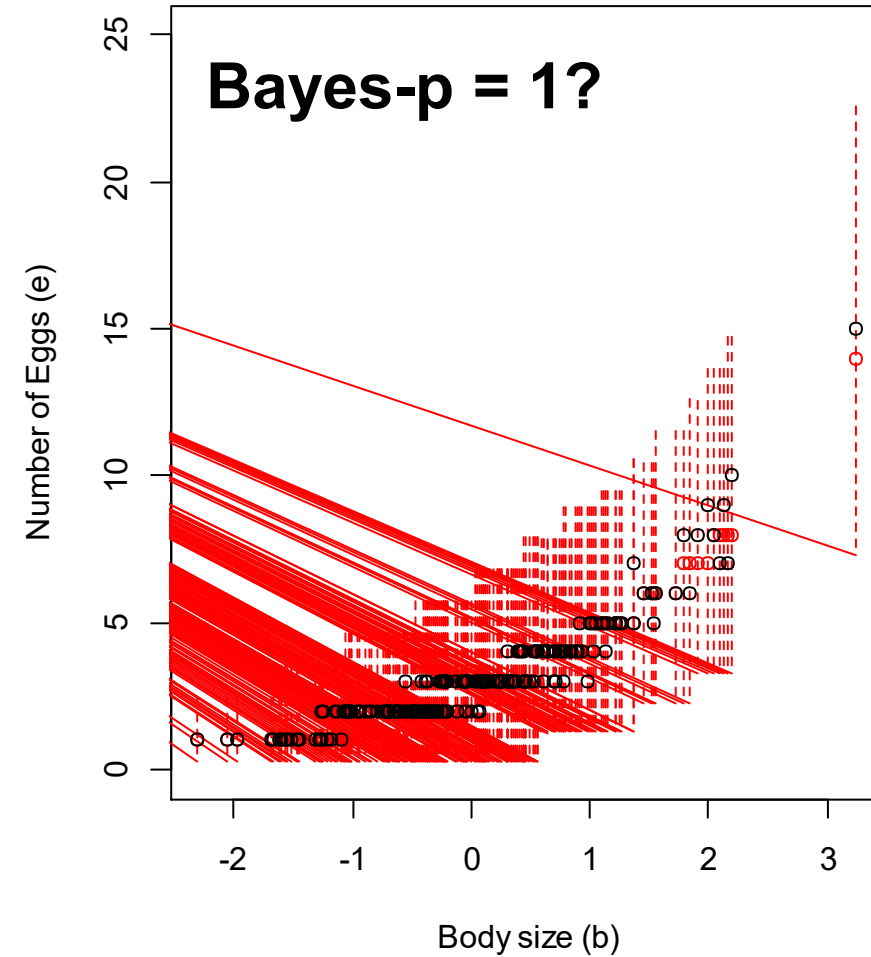
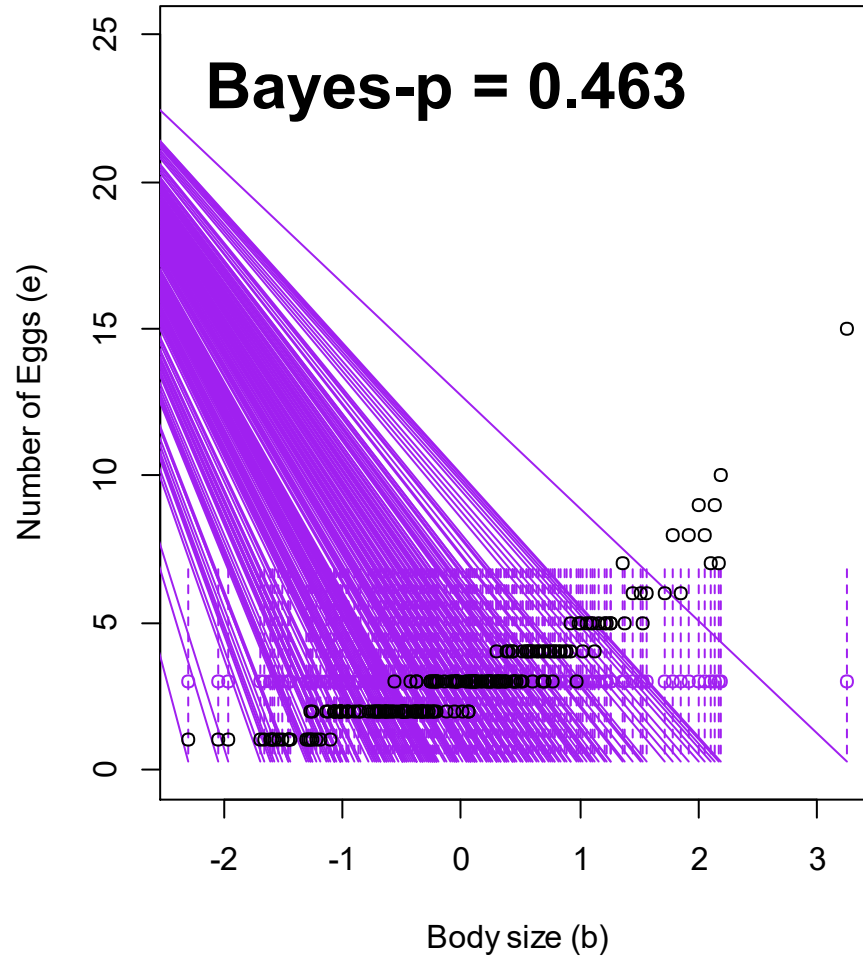


Oh, for goodness sake!!



Bayes-p > 0.5 means real data fit better than simulated, and vice versa

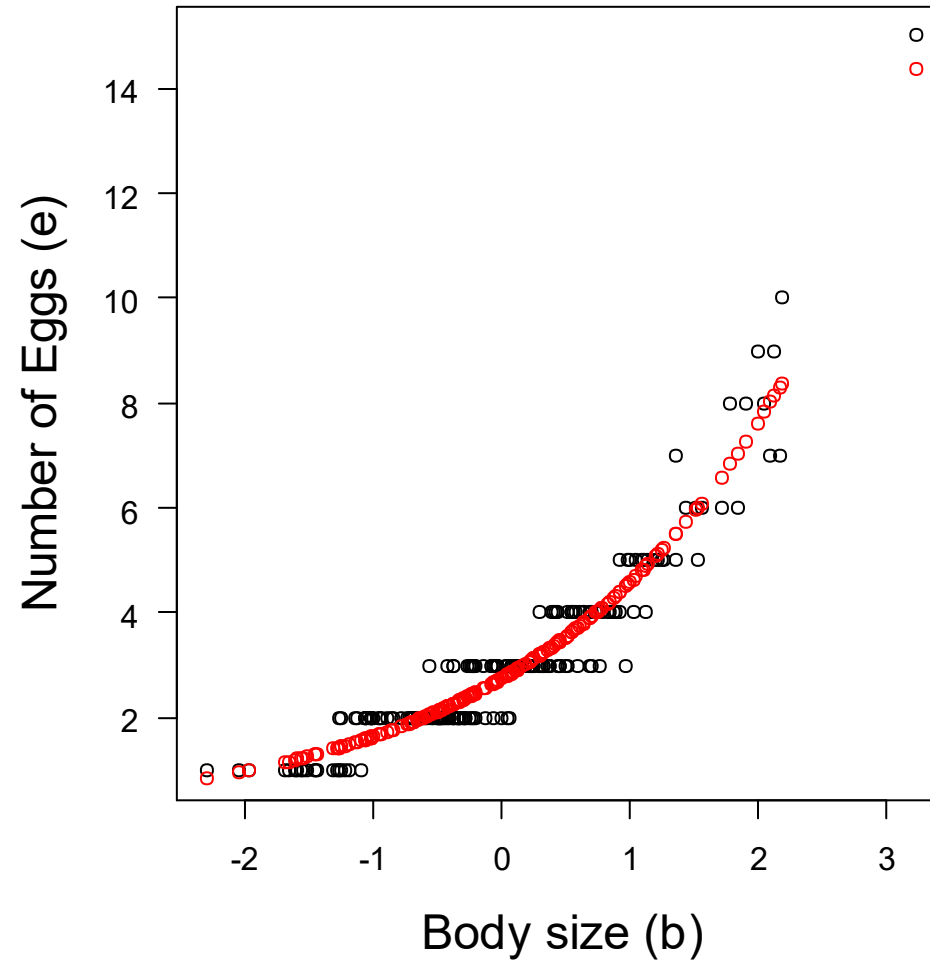
Oh, for goodness sake!!



Bayes-p > 0.5 means real data fit better than simulated, and vice versa

Is this model 'bad'?

It certainly doesn't fit the data well



Let's dig into the code and find out why!?

It will be fun.

A take-home message here is that poor model fit doesn't necessarily mean a 'bad' model. Rather, it may just mean a bit more thought is necessary for addressing error?

